

[12pt]article

sbc-template graphicx,url [utf8]inputenc [brazil]babel float booktabs hyperref

Soma de Vetores em CUDA: Uma Comparao de Desempenho entre CPU e GPU

Wsio de Arajo Filho¹

Instituto Federal Goiano – Campus Ipor

document

abstract This paper presents a performance comparison between sequential (CPU) and parallel (GPU) implementations of vector addition, a classic introductory problem to CUDA programming. Both implementations were written in C/CUDA and executed on an NVIDIA T4 GPU. Execution times were measured for four vector sizes: 1,000, 100,000, 1,000,000, and 10,000,000 elements, with each test repeated five times. The results show that the GPU outperforms the CPU for sufficiently large vectors, achieving a speedup of up to 53x at 10 million elements. For small vectors, the CPU is faster due to the overhead of data transfer and kernel launch in the GPU.